

Fraol Batole

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SUMMARY

PhD student building reliable, verifiable LLM agent systems: runtime verification for agent actions, multi-agent code analysis, and program-analysis techniques for AI-generated and AI-assisted code. First-author publications at ASE, ICSE, and IEEE TSE; shipped an IntelliJ refactoring plugin built with JetBrains. Seeking Research Scientist / Applied Scientist roles applying program analysis and empirical methods to make LLM agents and coding tools more reliable.

EXPERIENCE

Microsoft — GitHub Copilot Team

Jul 2026 – Oct 2026

Student Researcher (Contract), Manager Dr. Gaurav Mittal

Remote/Seattle, WA

- Conduct applied research on coding agents within the GitHub Copilot team.

Tulane University — Laboratory for Software Engineering

Aug 2024 – Present

Graduate Research Assistant, advised by Prof. Hridesh Rajan

New Orleans, LA

- **Thesis: Toward Reliable and Verifiable AI Agent Systems** — research on making LLM agents and AI-assisted software trustworthy, combining runtime verification, static/program analysis, and large-scale empirical study to detect and prevent agent failures; five first-author papers at ASE, ICSE, and IEEE TSE.

JetBrains

2023 – 2025

Research Collaborator, mentored by Dr. Danny Dig and Dr. Tien N. Nguyen

Remote

- Built LLM-based refactoring tools for the IntelliJ IDEA platform, starting with MM-Assist, an automated Move Method plugin combining LLMs, RAG, and hallucination filtering that reached 73% accuracy — a 4× improvement over prior tools (ICSME '25) — then extending the approach to broader agentic refactoring for IntelliJ.

Polytechnique Montréal

May 2025 – Aug 2025

Research Intern, advised by Prof. Foutse Khomh

Montreal, Canada

- Investigated safety and robustness of LLM-based AI agents, developing detection and mitigation strategies for agent misbehavior; findings directly informed RefineAct.

TECHNICAL SKILLS

Programming Languages: Python, Java, Prolog

ML / LLM Frameworks: PyTorch, TensorFlow/Keras, Hugging Face Transformers

Agents & Tooling: LangChain, LangGraph, CrewAI, multi-agent frameworks, Model Context Protocol (MCP)

Program Analysis / SE: Static analysis, dynamic analysis, tpestate analysis, runtime verification

Cloud / Infrastructure: Google Cloud Platform, Microsoft Azure, HPC clusters (ACCESS/XSEDE)

EDUCATION

Tulane University — Ph.D., Computer Science

Aug 2024 – Present

New Orleans, LA | GPA: 4.0/4.0 | Advisor: Prof. Hridesh Rajan

Iowa State University — M.Sc., Computer Science

May 2024

Ames, IA | GPA: 3.62/4.0

Addis Ababa Science and Technology University — B.Sc., Software Engineering

Jun 2018

Addis Ababa, Ethiopia | GPA: 3.75/4.0 | Magna Cum Laude

SELECTED PUBLICATIONS (FIRST-AUTHOR)

1. Fraol Batole, Foutse Khomh, and Hridesh Rajan. "Automatic Runtime Verification of LLM Agent Actions." *ASE '26*.
2. Fraol Batole, Ruchira Manke, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "Typestate-Based Fault Localization of API Usage Violations in Deep Learning Programs." *IEEE Transactions on Software Engineering (TSE '26)*.
3. Fraol Batole, David O'Brien, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "An LLM-Based Agent-Oriented Approach for Automated Code Design Issue Localization." *ICSE '25*.
4. Fraol Batole, Smit Patel, Aashish Yadavally, Tien N. Nguyen, and Hridesh Rajan. "CAPS: Code Abstraction Pre-training Strategy for Large Language Models." *Under submission*.
5. Fraol Batole, Ruchira Manke, David O'Brien, Steven Vo, Tien N. Nguyen, and Hridesh Rajan. "A Comprehensive Study on LLM-Based Agent Bug Characteristics." *Under submission*.

Plus 7 co-authored publications at FSE, ICSE, and ICSME — full list at fraolbatole.github.io.

HONORS, AWARDS & SERVICE

- Distinguished Reviewer Award (Junior PC), MSR '26 (Apr 2026).
- NSF-Funded Advanced Computing Credits: 1.5 million credits awarded (2024, 2025).
- Google Cloud Research Credits (\$12,000) and Google Cloud Research Innovator '23.
- Reviewer / Program Committee: MSR '26 (Junior PC), TOSEM '26, ASE Journal '25, Deep Learning for Code Workshop @ NeurIPS '25.

ADDITIONAL EXPERIENCE

Iowa State University — Graduate Teaching Assistant

Jan 2021 – Aug 2024

- Taught programming principles, debugging, and test-case development to 80+ students; adapted pedagogy for diverse learning styles.

MED Innovation — Software Engineer

Aug 2019 – Nov 2019

- Integrated a CRM system for a cross-functional team using Spring Boot and Vaadin.